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Original Research Article

Effect of multivitamin-mineral supplementation versus placebo on cognitive function: results from the clinic subcohort of the COcoa Supplement and Multivitamin Outcomes Study (COSMOS) randomized clinical trial and meta-analysis of 3 cognitive studies within COSMOS

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A B S T R A C T

Background: Longer effects of multivitamin-mineral (MVM) supplementation on late-life cognitive function remain untested using in-person, detailed neuropsychological assessments. Furthermore, insufficient evidence exists for healthcare providers to recommend daily MVM supplements to prevent cognitive decline.

Objectives: This study aimed to test MVM effects on cognitive change using in-person, detailed neuropsychological assessments and conduct a meta-analysis within COSMOS (COcoa Supplement and Multivitamin Outcomes Study) cognitive substudies for a robust evaluation of MVM effects on cognition.

Methods: COSMOS is a 2 × 2 factorial trial of cocoa extract (500 mg flavanols/d) and/or a daily MVM supplement for cardiovascular disease and cancer prevention among 21,442 United States adults aged ≥60 y. There were 573 participants in the clinic subcohort of COSMOS (that is, COSMOS-Clinic) who completed all cognitive tests administered at baseline. For the meta-analysis, we included nonoverlapping participants across 3 COSMOS cognitive substudies: COSMOS-Clinic (n = 573); COSMOS-Mind (n = 2158); COSMOS-Web (n = 2472).

Results: In COSMOS-Clinic, we observed a modest benefit of MVM compared with placebo on global cognition over 2 y {mean difference [95% confidence interval (CI)] = 0.06 SD units (SU) (−0.003, 0.13)}, with a significantly more favorable change in episodic memory [mean difference (95% CI) = 0.12 SU (0.002, 0.23)] but not in executive function or attention [mean difference (95% CI) = 0.04 SU (−0.04, 0.11)]. The meta-analysis of COSMOS substudies showed clear evidence of MVM benefits on global cognition [mean difference (95% CI) = 0.07 SU (0.03, 0.11); P = 0.0009] and episodic memory [mean difference (95% CI) = 0.06 SU (0.03, 0.10); P = 0.0007]; the magnitude of effect on global cognition was equivalent to reducing cognitive aging by 2 y.

Abbreviations: 3MS, Modified Mini-Mental State; ADRD, Alzheimer's disease and related dementias; CCI, Center for Clinical Investigation; CE, cocoa extract; CERAD, Consortium to Establish a Registry for Alzheimer's Disease; CI, confidence interval; COSMOS, COcoa Supplement and Multivitamin Outcomes Study; CVD, cardiovascular disease; EBMT, East Boston Memory Test; IU, international unit; MVM, multivitamin-mineral; PHS II, Physicians' Health Study II; RCT, randomized clinical trial; STIDA, Structured Telephone Interview for Dementia Assessment; SU, SD units; τ^2, tau squared; TMT, Trail Making Test.

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Conclusions: In COSMOS-Clinic, daily MVM supplementation leads to a significantly more favorable 2-y change in episodic memory. The meta-analysis within COSMOS cognitive substudies indicates that daily MVM significantly benefits both global cognition and episodic memory. These findings within the COSMOS trial support the benefits of a daily MVM in preventing cognitive decline among older adults.
This trial was registered at COSMOS—clinicaltrials.gov as NCT02422745, at COSMOS-Mind—clinicaltrials.gov as NCT03035201, and at COSMOS-Web—clinicaltrials.gov as NCT04582617.

Keywords: multivitamin-multimineral supplementation, cognition, randomized clinical trial, meta-analysis, geriatric

Introduction

By 2060, nearly 1 in 4 Americans will be aged ≥65 y—an age group at elevated risk of cognitive decline and Alzheimer's disease and related dementias (ADRD) [1,2]. Studies have shown that even modest deficits in cognitive function during aging can predict worsened cognitive decline and an increased risk of ADRD in late life [3,4]; these early signs of cognitive decline may be delayed through a safe, efficacious, and broadly applicable intervention, such as a multivitamin-mineral (MVM) supplement. Although a few long-term randomized clinical trials (RCTs) have tested MVM supplements on cognitive change in older adults [5–7], effects remain untested using in-person detailed neuropsychological assessments, capturing subtle changes across multiple cognitive domains.

A typical broad-based MVM supplement contains most or all essential vitamins and minerals at lower levels than individual supplements to target multiple biologic pathways to support cognitive health [8–10]. Prior RCTs examining single or smaller combinations of nutrients have shown no or limited benefits on late-life cognition [9–13]. Only 3 long-term RCTs (>12 mo) have tested a daily broad-based MVM on cognitive change in older adults. In the Physicians' Health Study II (PHS II), telephone-based cognitive assessments began after an average duration of 2.5 y following randomization to intervention groups and found no MVM benefits on cognition [5]. In the COSMOS (COcoa Supplement and Multivitamin Outcomes Study), initial cognitive assessments took place before randomization for COSMOS-Mind (n = 2262; telephone-based) [6] and COSMOS-Web (n = 3960; computer-based) [7] and found evidence of MVM benefits on cognitive function. However, in-person, detailed neuropsychological assessments, with the potential to measure subtle changes across multiple cognitive domains, have not been previously conducted to confirm the effects of MVM supplementation on cognitive function among older adults.

Therefore, we tested the effects of MVM supplementation on cognitive change in the clinic subcohort of 573 COSMOS participants (that is, COSMOS-Clinic) who underwent in-person detailed neuropsychological assessments at baseline and 2 y. Because COSMOS is unique in integrating 3 separate cognitive ancillary studies, all of which had prerandomization and follow-up neuropsychological assessments within the broader parent trial, a meta-analysis among nonoverlapping participants (n > 5000) across COSMOS-Clinic, COSMOS-Mind, and COSMOS-Web was conducted to evaluate the strength of evidence regarding the effects of an MVM on cognition with potential clinical and public health relevance.

Methods

COSMOS trial

COSMOS is a recently completed randomized, double-blind, placebo-controlled, 2 × 2 factorial design testing cocoa extract (CE; Mars Edge; 500 mg flavanols/d, including 80 mg (−)-epicatechin) and/or a daily MVM supplement (Haleon; Centrum Silver®) for cardiovascular disease (CVD) and cancer prevention in 21,442 United States adults, aged ≥60 y; protocol details and main findings are published elsewhere [14–16]. Randomization was performed using a computer-generated sequential list of random allocations for the 4 treatment combinations. The random allocation sequence was created using SAS version 9.4 (SAS Institute) and was stratified by sex, age (5-y bins), and recruitment source (that is, mass mailings to Women's Health Initiative Extension Study participants, or mass mailings to participant recruitment pools from large studies led by Brigham and Women's Hospital). All trial investigators, examiners, and participants were masked to treatment group assignment. Randomization occurred from June 2015 through March 2018; randomized treatment ended on 31 December, 2020, and this was the final date of follow-up.

The exclusion criteria for the COSMOS trial were as follows: *1*) history of myocardial infarction or stroke (individuals with other cardiovascular events, including transient ischemic attack, congestive heart failure, coronary artery bypass graft, angioplasty, and stent were permitted); *2*) diagnosis of cancer except for nonmelanoma skin cancer within the past 2 y; *3*) other serious illnesses precluding participation; *4*) unwillingness to forego the use of CE, multivitamins, vitamin D (>1000 international unit (IU)/d), or high-dose calcium (>1200 mg/d) supplements during the trial; *5*) extreme sensitivity to caffeine; *6*) <75% compliance to study pills during the study period; and *7*) inability to communicate in English. Regarding compliance, participants answered study pill adherence questions (that is, number of days per month taking study pills from their monthly calendar packs) in questionnaires every 6 mo [14].

COSMOS-clinic cognitive substudy

A subcohort of 603 participants from the parent COSMOS trial who lived within driving distance of the Center for Clinical Investigation (CCI) at Brigham and Women's Hospital (that is, COSMOS-Clinic) were invited to complete detailed, in-person health evaluations at CCI, which included 45-min neuropsychological evaluations at baseline and 2 y [14]. All participants provided written informed consent, and study approvals were obtained from the Institutional Review Board at Mass General Brigham. COSMOS-Clinic cognitive substudy analyses were prespecified as tertiary aims on the clinicaltrials.gov registration page of COSMOS (NCT02422745).

COSMOS-clinic cognitive substudy procedures

Study procedures followed a protocol developed before clinic enrollment and randomization (see Supplemental Methods A). Of 603 COSMOS participants enrolled in the COSMOS-Clinic subcohort, 602 were interested in neuropsychological evaluations and completed hearing screening [17] and the Clock-in-the-Box test [18], a rapid (1–2 min) cognitive screening test, followed by the Modified Mini-Mental State (3MS) [19], a measure of general cognition. Participants without serious hearing impairment were administered these additional neuropsychological measures: *1*) Consortium to Establish a Registry

for Alzheimer's Disease [(CERAD) immediate total learning, delayed recall, and recognition] [20,21], *2*) immediate and delayed recall trials of the East Boston Memory Test (EBMT) [22], *3*) 2 category fluency tests (naming animals and vegetables) [20], *4*) Trail Making Tests (TMTs) A and B [23], and *5*) Digit Span Backwards [24]. COSMOS-Clinic neuropsychological assessments were conducted at baseline and follow-up an average of 2 y later (interquartile range: 2.0–2.1 y).

COSMOS-clinic: ascertainment of primary and secondary outcomes

In COSMOS-Clinic, our primary outcome was the global cognition composite (averaging mean standardized (z) scores of 11 tests: 3MS; CERAD immediate total learning, delayed recall, and recognition; immediate and delayed EBMT recall trials; 2 category fluency tests; TMT-A and TMT-B; Digit Span Backwards), with higher z-scores reflecting better performance.

Secondary outcomes were composites of episodic memory (averaging z-scores of 4 tests: EBMT immediate and delayed recall trials, and CERAD immediate total learning and delayed recall) and executive function/attention (averaging z-scores of 5 tests: 2 category fluency tests, TMT-A, TMT-B, and Digit Span Backwards). We standardized follow-up composite scores using fixed baseline means and SDs of the respective cognitive measures.

Meta-analysis: description of COSMOS cognitive substudies

In the meta-analysis, we included nonoverlapping participants across the 3 COSMOS cognitive substudies: *1*) COSMOS-Clinic (n = 573; in-person cognitive assessments over 2 y); *2*) COSMOS-Mind (n = 2158; annual telephone-based cognitive assessments for 3 y) [6]; and *3*) COSMOS-Web (n = 2472; annual computer-based cognitive assessments for 3 y) [7]. MVM findings of COSMOS-Mind and COSMOS-Web are published elsewhere [6,7]. We prioritized COSMOS-Clinic, followed by COSMOS-Mind, and then COSMOS-Web study samples to exclude overlapping participants across these studies. Our selection favored a study with a smaller sample size when choosing data of overlapping participants across these studies.

Meta-analysis: coprimary cognitive outcomes

Global cognition and episodic memory were prespecified as the coprimary cognitive outcomes for this meta-analysis because of their relevance to age-related cognitive decline and ADRD. Global cognition reflects overall ability across multiple cognitive domains; episodic memory declines with aging and is typically the primary cognitive domain affected by AD, the leading cause of dementia [25].

Supplemental Table 1 provides a list of neuropsychological measures used in each COSMOS cognitive substudy to assess meta-analysis outcomes. The COSMOS-Clinic and COSMOS-Mind cognitive substudies had parallel approaches to assessing global cognition and episodic memory outcomes. The COSMOS-Web was specifically designed to examine the treatment effects on hippocampus-mediated episodic memory [7]. As a result, COSMOS-Web did not report global cognition as a prespecified cognitive endpoint. Thus, the meta-analysis of global cognition included nonoverlapping participants from the COSMOS-Clinic and COSMOS-Mind studies. Regarding the meta-analysis of episodic memory, we included nonoverlapping participants across 3 COSMOS cognitive substudies.

Statistical analyses

COSMOS-clinic: main analyses

The study was designed to have >80% power to detect an effect size of 0.12 SD units (SU) or greater in the global cognition composite score comparing MVM and placebo groups, with a planned sample size of 500 and assumed 10% loss to follow-up (see Supplemental Methods B).

The analysis for the effect of MVM supplementation compared with placebo included all randomly assigned participants, regardless of pill compliance, participation in the year 2 follow-up, and scores from all cognitive assessments (baseline or year 2); 573 participants who completed all neuropsychological tests at baseline made up the sample (see CONSORT diagram, Figure 1). Descriptive characteristics of participants were compared between MVM and placebo groups. To examine the mean difference in change in global cognition over time between the MVM and placebo groups, we used repeated measures models over a 2-y follow-up time with an unstructured covariance matrix to incorporate the within-participant correlation. Models were adjusted for the prespecified design variables: age, sex, randomization to CE, and CE effect over time (that is, time-x-CE). We estimated the mean difference between the MVM and placebo groups in change in global cognition score over time using a time-x-MVM interaction. Regarding model assumptions, the averaged z-score cognitive outcomes met assumptions for normality and missing data were considered missing at random; as the repeated measures models can handle missing data, all participants contributed to the analysis at one or both time points. Models were fitted by maximum likelihood and we used Wald tests for statistical testing [26]. For secondary outcomes, we constructed repeated measures models with the abovementioned model specifications and covariates adjustment.

COSMOS-clinic: subgroup analyses

We addressed whether the effect of MVM supplementation compared with placebo on global cognitive change over 2 y was modified by design variables or subgroups selected *a priori* on the basis of the importance in cognitive decline and ADRD [27,28]: age (<70 or ≥70 y); sex; self-identified racial and ethnic group {non-Hispanic White, or other racial and ethnic groups [Black or African American, Hispanic (not Black or African American)], Asian, Native Hawaiian or other Pacific Islander, American Indian or Alaska Native, and multiple or unknown race and/or unknown ethnicity; see details in Table 1}; CE randomization group; healthy diet adherence level (tertiles of the Alternative Healthy Eating Index-2010 score) [29]; baseline general cognition score (3MS tertiles); presence of subjective cognitive concerns [0 or ≥1 points on the Structured Telephone Interview for Dementia Assessment (STIDA)] [30]; body mass index (<25, 25–29, ≥30 kg/m^2); history of CVD (on the basis of self-report of transient ischemic attack, congestive heart failure, coronary artery bypass graft, angioplasty, or stent), hypertension, type 2 diabetes, and depression.

For all COSMOS-Clinic analyses, tests were two-sided, and a P value < 0.05 was used for statistical significance. The analyses of secondary outcomes and subgroups were considered hypothesis generating and were not adjusted for multiple comparisons; thus, results should be interpreted with caution.

Meta-analysis of prespecified outcomes within COSMOS cognitive substudies

We performed a meta-analysis for the effect of MVM supplementation on global cognition and episodic memory using

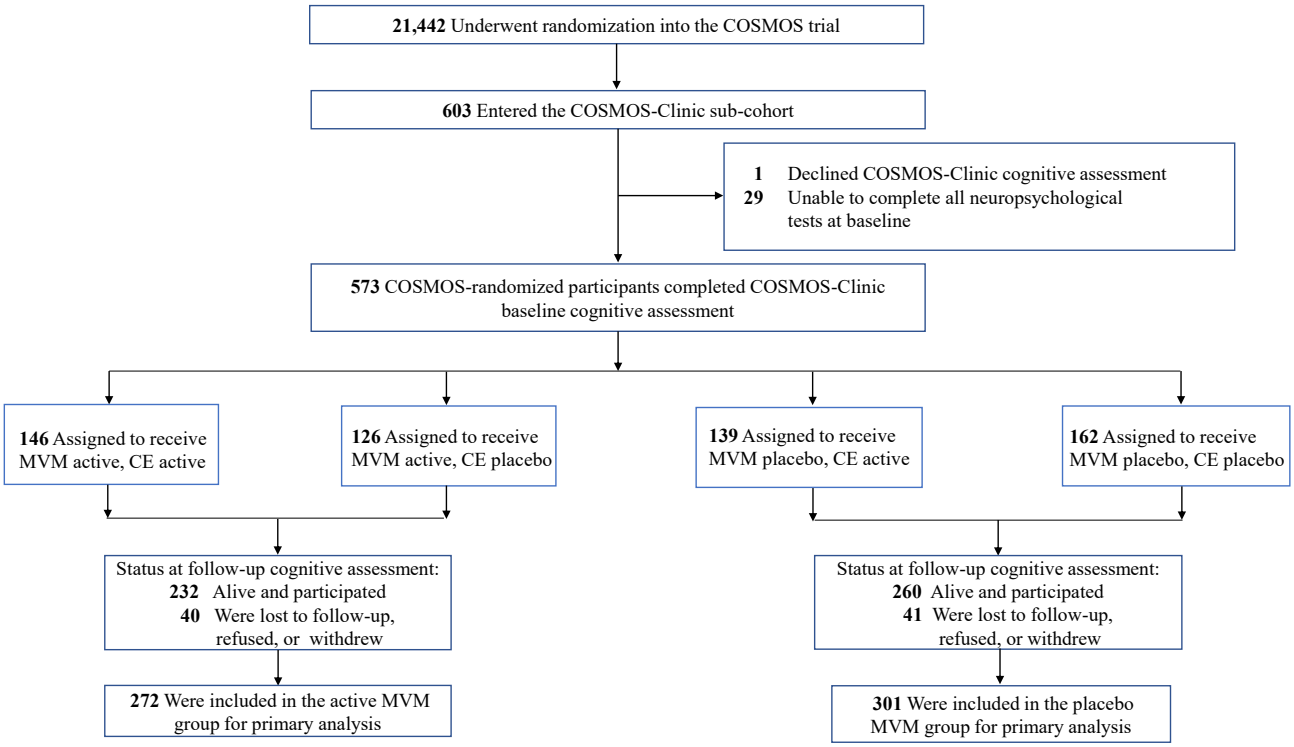

FIGURE 1. Flow of participants in the COSMOS-Clinic cognitive substudy. CE, cocoa extract; COSMOS, the COcoa Supplement and Multivitamin Outcomes Study [14–16]; MVM, multivitamin-mineral.

random-effects models. The outcomes were mean standardized (z) scores, relative to baseline, in each cognitive substudy; the study-specific estimates were pooled, and a summary effect estimate [95% confidence interval (CI)] and P value were reported. Heterogeneity across substudies (using the Q statistic) and between-study variance [using tau squared (τ^2)] were examined using the %METAANAL macro (DerSimonian–Laird method) [31,32]. Tests were two-sided; for an α-level of 0.05 with 2 coprimary outcomes, $P < 0.025$ was used to indicate statistical significance for each coprimary outcome after Bonferroni correction [33].

All analyses were performed using SAS version 9.4 (SAS Institute) and R version 4.1.1 (R Foundation for Statistical Computing).

Results

Of the 603 COSMOS-Clinic participants, 573 completed all neuropsychological tests at baseline (Figure 1). Among these 573 participants, 81 (14.1%) were lost to follow-up, refused follow-up cognitive assessment, or withdrew from the study before the follow-up cognitive assessment; 492 of 573 (85.9%) completed all neuropsychological assessments at baseline and 2 y. Among participants who returned study pill adherence questionnaires (98% of this sample), 95.5% in the MVM group, and 95.3% in the placebo group reported adequate study pill adherence (that is, taking ≥75% study pills per month) at year 1. At 2 y, 90.9% of the MVM group and 90.0% of the placebo group reported adequate study pill adherence. This report includes results regarding the effects of MVM on cognition in the COSMOS-Clinic cognitive substudy; results regarding the effects of CE on cognition are described elsewhere [34].

COSMOS-clinic: baseline characteristics

There were 272 participants randomly assigned to MVM, and 301 were randomly assigned to placebo. The mean (SD) age was 69.6 (5.3) y, females comprised 49.2%, and 37.9% reported prerandomization use of multivitamins (Table 1). Descriptive characteristics were comparable across treatment groups.

COSMOS-clinic: main analyses

Participants randomly assigned to MVM compared with placebo showed a modest benefit in global cognition over 2 y; the mean difference in change in global cognition score, relative to baseline, was 0.06 SU (95% CI: −0.003, 0.13) (Table 2).

Regarding secondary outcomes, MVM supplementation compared with placebo led to a statistically significant more favorable change in episodic memory; the mean difference in change in episodic memory score, comparing the MVM and placebo groups, was 0.12 SU (95% CI: 0.002, 0.23) (Table 2). To assess the clinical importance of findings, we compared the observed estimate of the treatment effect with the estimate within the multivariable model for a 1-y increase in participant age, because age is the most robust predictor of cognitive decline. The estimate for the association between a 1-y difference in age at baseline and a 2-y change in episodic memory was −0.025 SU; this translates to the effect of a 2-y MVM supplementation that was equivalent to 4.8 y less aging in episodic memory than placebo. In contrast, there was no MVM benefit for executive function or attention over 2 y in this sample.

COSMOS-clinic: subgroup analyses

We did not observe significant interactions by selected subgroups for the effect of MVM supplementation compared with placebo on

TABLE 1
Characteristics of COSMOS-Clinic cognitive substudy participants at baseline

Characteristic	No. (%) of participants[1]	
	MVM group (n = 272)	Placebo group (n = 301)
Age, mean (SD)	69.3 (5.1)	69.8 (5.5)
Sex, n (%)		
Male	139 (51.1)	152 (50.5)
Female	133 (48.9)	149 (49.5)
Randomization to CE supplement		
Active group	146 (53.7)	139 (46.2)
Placebo group	126 (46.3)	162 (53.8)
Self-reported racial and ethnic group,[2] n (%)		
Non-Hispanic White	267 (98.2)	286 (95.0)
Other racial and ethnic groups	5 (1.8)	15 (5.0)
Education, n (%)	(n = 268)	(n = 297)
≤College	107 (39.9)	151 (50.8)
Post-college	161 (60.1)	146 (49.2)
BMI, median (IQR), (kg/m^2)	27.1 (24.1–30.4)	27.3 (24.5–30.6)
Leisure-time physical activity, median (IQR), MET (h/wk)	21.5 (6.9–36.1)	19.0 (9.1–37.3)
Adherence to a healthy diet, AHEI-2010 (points), n (%)	(n = 246)	(n = 275)
Lowest tertile (15.5–38.5)	68 (27.6)	101 (36.7)
Middle tertile (39.5–47.5)	84 (34.2)	90 (32.7)
Highest tertile (48.5–72.5)	94 (38.2)	84 (30.6)
Smoking status, n (%)	(n = 269)	(n = 298)
Never	147 (54.7)	164 (55.0)
Past or current	122 (45.4)	134 (45.0)
Frequency of alcohol use, n (%)	(n = 256)	(n = 286)
Never or rarely	54 (21.1)	69 (24.1)
Monthly	20 (7.8)	13 (4.6)
Weekly	99 (38.7)	114 (39.9)
Daily	83 (32.4)	90 (31.5)
Regular use of multivitamin supplement, n (%)	105 (38.6)	112 (37.3)
History of hypertension, n (%)	125 (46.0)	157 (52.2)
History of CVD,[3] n (%)	16 (6.0)	19 (6.4)
History of diabetes, n (%)	25 (9.2)	38 (12.7)
History of high-cholesterol medication, n (%)	109 (40.2)	116 (38.7)
History of depression, n (%)	80 (29.5)	77 (25.8)
Baseline cognitive outcomes, SU, mean (SD)		
Primary outcome		
Global cognition composite[4]	0.02 (0.57)	−0.02 (0.54)
Secondary outcomes		
Episodic memory composite[5]	0.01 (0.74)	−0.01 (0.72)
Executive attention or function composite[6]	0.04 (0.65)	−0.04 (0.66)
Presence of subjective cognitive complaints,[7] n (%)		
Yes	86 (31.6)	98 (32.6)
No	186 (68.4)	203 (67.4)

Abbreviations: 3MS, Modified Mini-Mental State Examination [19]; AHEI, alternative healthy eating index; CE, cocoa extract; CERAD, Consortium to Establish a Registry for Alzheimer's Disease [20,21]; COSMOS, COcoa Supplement and Multivitamin Outcomes Study; CVD, cardiovascular disease; IQR, interquartile range; MET, metabolic equivalent of task; MVM, multivitamin-mineral; SU, SD units.

[1] Unless otherwise indicated. The percentages may not sum to 100 because of rounding.

[2] In the COSMOS prerandomization questionnaire, participants were presented with the following categories of race and asked to check all that apply: American Indian or Alaska Native; Black or African American; Asian; Native Hawaiian or other Pacific Islander; White; Unknown. Regarding ethnicity, participants were asked to select one of these options: Hispanic; non-Hispanic; Unknown. On the basis of the responses, we derived a binary variable for self-reported racial and ethnic background, with categories of non-Hispanic White and other racial and ethnic groups (Black or African American; Hispanic (not Black or African American); Asian; Native Hawaiian or other Pacific Islander; American Indian or Alaska Native; multiple races or unknown race and/or unknown ethnicity).

[3] On the basis of the self-report of transient ischemic attack, congestive heart failure, coronary artery bypass graft, angioplasty, or stent.

[4] Global cognition composite was obtained by averaging z-scores of 11 tests: 3MS [19], immediate and delayed recalls of the East Boston Memory Test [22], category fluency tests (naming animals and vegetables) [20], tests from CERAD (immediate learning, delayed recall, recognition) [20,21], Trail Making Tests A and B [23], and Digit Span Backwards [24].

[5] Episodic memory composite was obtained by averaging z-scores of 4 tests: immediate and delayed recalls of the East Boston Memory Test [22], and CERAD immediate learning and delayed recall tests [20,21].

[6] Executive function or attention composite was obtained by averaging z-scores of 5 tests: category fluency tests (naming animals and vegetables) [20], Trail Making Tests A and B [23], and Digit Span Backwards [24].

[7] Subjective cognitive complaints were ascertained using the Structured Telephone Interview for Dementia Assessment (STIDA) [30].

TABLE 2
Adjusted mean difference in change in cognitive outcomes over a 2-y follow-up, comparing MVM and placebo groups[1]

	MVM group		Placebo group		Adjusted mean difference (95% CI) for change over a 2-y follow-up, SU
	N	Mean (SE)[2]	*N*	Mean (SE)[2]	
Primary outcome					
Global cognition[3]					
Baseline	272	0.02 (0.03)	301	−0.01 (0.03)	
Year 2	232	0.19 (0.03)	260	0.09 (0.03)	0.06 (−0.003, 0.13)
Secondary outcomes					
Episodic memory[4]					
Baseline	272	0.01 (0.04)	301	−0.01 (0.04)	
Year 2	232	0.36 (0.04)	260	0.23 (0.04)	0.12 (0.002, 0.23)
Executive function or attention[5]					
Baseline	272	0.03 (0.04)	301	−0.03 (0.04)	
Year 2	232	0.05 (0.04)	260	−0.05 (0.04)	0.04 (−0.04, 0.11)

Abbreviations: 3MS, Modified Mini-Mental State Examination [19]; CE, cocoa extract; CERAD, Consortium to Establish a Registry for Alzheimer's Disease [20, 21]; CI, confidence interval; MVM, multivitamin-mineral; SU, SD units.

[1] Results are from the repeated measures model; all participants contributed to the repeated measure analysis at one and/or both time points. Models were controlled for baseline age, sex, randomization to the CE group, and CE effect over time (that is, CE-x-time).

[2] Mean and SEs (in SU) are calculated from a multivariable model adjusted by age, sex, randomization to the CE group, and CE effect over time (that is, CE-x-time). The analyses of secondary outcomes were not adjusted for multiple comparisons.

[3] Global cognition composite was obtained by averaging *z*-scores of 11 tests: 3MS [19], immediate and delayed recalls of the East Boston Memory Test [22], category fluency tests (naming animals and vegetables) [20], tests from CERAD (immediate learning, delayed recall, recognition) [20,21], Trail Making Tests A and B [23], and Digit Span Backwards [24].

[4] Episodic memory composite was obtained by averaging *z*-scores of 4 tests: immediate and delayed recalls of the East Boston Memory Test [22], and CERAD immediate learning and delayed recall tests [20,21].

[5] Executive function or attention composite was obtained by averaging *z*-scores of 5 tests: category fluency tests (naming animals and vegetables) [20], Trail Making Tests A and B [23], and Digit Span Backwards [24].

global cognitive change over 2 y (Figure 2); furthermore, there was no significant interaction between the MVM and CE interventions for global cognition (P-interaction = 0.46).

Meta-analysis of prespecified outcomes within COSMOS cognitive substudies

The meta-analysis results for global cognition (Figure 3) show that participants randomly assigned to the MVM group compared with placebo had a statistically significant more favorable change in global cognition over time (mean difference (95% CI): 0.07 (0.03, 0.11) SU; $P = 0.0009$); findings remained significant after Bonferroni correction. The overall effect was not heterogeneous across studies (P-heterogeneity = 0.88), with no between-study variance ($\tau^2 = 0\%$).

The meta-analysis results for episodic memory (Figure 3) show that participants randomly assigned to the MVM group compared with placebo had a statistically significant more favorable change in episodic memory over time [mean difference (95% CI): 0.06 (0.03, 0.10) SU; $P = 0.0007$]; findings remained significant after Bonferroni correction. The overall effect was not heterogeneous across studies (P-heterogeneity = 0.61), with no between-study variance ($\tau^2 = 0\%$).

Regarding the clinical importance of our meta-analysis findings: the magnitude of the effect of MVM supplementation compared with placebo on global cognition score was equivalent on average to the difference observed between participants 2 y apart in age (that is, 2 y less cognitive aging).

Discussion

We observed a statistically significant beneficial effect of MVM supplementation compared with placebo on episodic memory in the well-characterized COSMOS-Clinic subcohort of 573 participants. Furthermore, our meta-analysis among over 5000 nonoverlapping participants across the 3 COSMOS cognitive substudies showed clear evidence of MVM benefits on both global cognition and episodic memory. Regarding clinical importance, the effect of daily MVM supplementation on global cognition over a duration of 2–3 y may translate to a reduction in cognitive aging by 2 y compared with placebo. Overall, our findings provide support for the benefits of an MVM supplement in preventing cognitive decline among older adults.

There is biologic plausibility for the findings, as combinations of vitamins and minerals interact with multiple biologic pathways that support cognitive health, and nutritional and micronutrient deficiencies in older adults may elevate risk of cognitive decline [8]. Prior RCTs testing single or limited combinations of vitamins or minerals on cognitive change have usually shown null results; however, short treatment durations (≤12 mo), lack of a baseline prerandomization cognitive assessment, the specific cognitive tests administered, insufficient statistical power, reliance on a post hoc analysis, or a combination of these factors limit cross-study comparisons [9–13]. In addition to the COSMOS-Clinic cognitive substudy, 3 other RCTs (that is, PHS II, COSMOS-Mind, and COSMOS-Web) [5–7] have investigated long-term (>12 mo) effects of a typical broad-based MVM on cognitive change among older adults. In COSMOS-Clinic, MVM supplementation led to better episodic memory scores over time compared with placebo; similar benefits were observed in COSMOS-Mind and COSMOS-Web studies [6,7]. Although COSMOS-Clinic had the smallest sample among the 3 COSMOS cognitive substudies, the estimate of the MVM effect on episodic memory was larger in the COSMOS-Clinic. The use of in-person assessments in COSMOS-Clinic, including the addition of tests from the CERAD, may have enhanced measurement precision.

Our meta-analysis of findings within COSMOS regarding the cognitive benefits of MVM supplementation among older adults has

Subgroup	No. of participants	Adjusted mean difference (95% CI) in change, SU	P−interaction[2]
Age, years			0.78
<70	334	0.05 (−0.03, 0.14)	
70+	239	0.07 (−0.03, 0.18)	
Sex			0.88
Female	282	0.06 (−0.03, 0.16)	
Male	291	0.06 (−0.03, 0.14)	
Self−reported racial and ethnic group[3]			0.07
Non−Hispanic White	553	0.05 (−0.01, 0.12)	
Other racial and ethnic groups	20	0.41 (0.04, 0.78)	
Randomization to CE supplement			0.46
Active group	285	0.09 (−0.004, 0.18)	
Placebo group	288	0.04 (−0.05, 0.13)	
AHEI−2010, points (range)			0.32
Lowest tertile (15.5−38.5)	169	0.09 (−0.04, 0.23)	
Middle tertile (39.5−47.5)	174	0.07 (−0.04, 0.19)	
Highest tertile (48.5−72.5)	178	0.01 (−0.09, 0.11)	
Body mass index, kg/m^2			0.83
Normal, <25.0	173	0.06 (−0.06, 0.19)	
Overweight, 25.0−29.9	230	0.09 (−0.01, 0.19)	
Obesity, 30.0+	170	0.04 (−0.07, 0.15)	
CVD[4]			0.90
Yes	35	0.06 (−0.15, 0.28)	
No	528	0.06 (−0.008, 0.13)	
Depression			0.28
Yes	157	0.11 (0.006, 0.22)	
No	413	0.04 (−0.04, 0.12)	
Hypertension			0.35
Yes	282	0.04 (−0.05, 0.12)	
No	291	0.09 (−0.004, 0.18)	
Diabetes			0.20
Yes	63	0.20 (0.001, 0.40)	
No	509	0.05 (−0.01, 0.12)	
General cognitive function (3MS),[5] points(range)			0.57
Lowest tertile (74.0−95.0)	188	0.12 (−0.01, 0.24)	
Middle tertile (96.0−98.0)	180	0.02 (−0.09, 0.14)	
Highest tertile (99.0−100.0)	205	0.06 (−0.05, 0.16)	
Presence of subjective cognitive complaints[6]			0.83
Yes	184	0.07 (−0.06, 0.19)	
No	389	0.05 (−0.02, 0.13)	

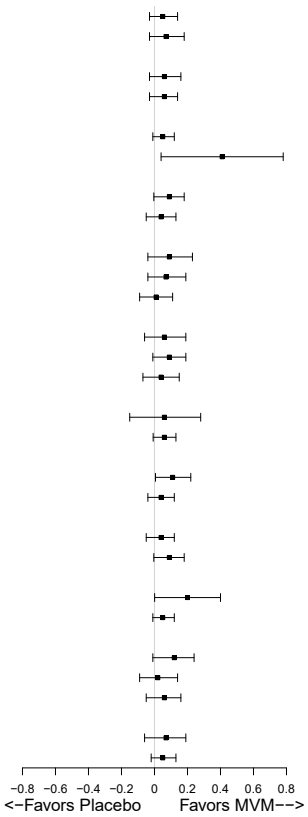

FIGURE 2. Adjusted mean difference in change in global cognition composite over a 2-y follow-up comparing MVM and placebo groups, according to prespecified subgroups.[1] 3MS, Modified Mini-Mental State [19]; AHEI, alternative healthy dietary index [29]; CE, cocoa extract; CI, confidence interval; CVD, cardiovascular disease; MVM, multivitamin-mineral; SU, SD units.

[1]Analyses were performed using repeated measures models adjusted for baseline age, sex, randomization to the CE group, and CE effect over time (that is, CE-x-time). All participants contributed to the repeated measure analysis at one and/or both time points. Adjusted mean differences (95% CI) between MVM and placebo groups in global cognition scores over a 2-y follow-up period are shown within subgroups. Analyses were not adjusted for multiple comparisons.

[2]*P*-interaction is from the 1-df test of the subgroup-x-treatment-x-follow-up time interaction term in the model.

[3]In the COSMOS prerandomization questionnaire, participants were presented with the following categories of race and asked to check all that apply: American Indian or Alaska Native; Black or African American; Asian; Native Hawaiian or other Pacific Islander; White; Unknown. Regarding ethnicity, participants were asked to select one of these options: Hispanic; non-Hispanic; Unknown. On the basis of the responses, we derived a binary variable for self-reported racial and ethnic background, with categories of non-Hispanic White and other racial and ethnic groups [Black or African American; Hispanic (not Black or African American); Asian; Native Hawaiian or other Pacific Islander; American Indian or Alaska Native; multiple race or unknown race and/or unknown ethnicity].

[4]On the basis of the self-report of transient ischemic attack, congestive heart failure, coronary artery bypass graft, angioplasty, or stent.

[5]For this subgroup analysis, we excluded the *z*-score of the general cognition test from the global cognition composite.

[6]Subjective cognitive complaints were ascertained using the Structured Telephone Interview for Dementia Assessment (STIDA) [30].

important implications for future clinical practice and scientific research. In clinical settings, MVM supplements merit greater consideration as a recommended treatment strategy to prevent cognitive decline among older adults without dementia at baseline. Also, in both PHS II and COSMOS, long-term MVM use is safe without notable adverse effects. Regarding implications for scientific research, further investigations should confirm findings among participants from more diverse backgrounds, with greater representation of the entire United States population. Additionally, future studies may incorporate ADRD biomarkers (for example, amyloid, tau), -omics (for example, epigenomics, microbiome, metabolomics), or genetics (for example, apolipoprotein E genotype) in an experimental framework to provide novel insights into how MVM supplements work to prevent cognitive decline and for whom they may be most effective.

Outside of the 3 COSMOS cognitive ancillaries (that is, COSMOS-Clinic, COSMOS-Mind, and COSMOS-Web), no other RCTs in the literature to date featured both prerandomization and follow-up neuropsychological assessments, a comprehensive MVM supplement intervention (containing ≥20 vitamins and minerals), intervention durations of 12 mo or longer, and sufficiently large sample sizes ($n \geq 500$). Results regarding the MVM intervention in COSMOS cognition ancillaries differ from earlier findings in PHS II ($n = 5947$; mean age: 71.6 y), in which no MVM benefits on cognitive change were observed over an average follow-up of 10 y [5]. However, important differences between the COSMOS and PHS II trials should be noted. First, COSMOS included both older males and females, whereas PHS II included only older males. Second, PHS II started cognitive testing, on average, 2.5 y after randomization. As a result,

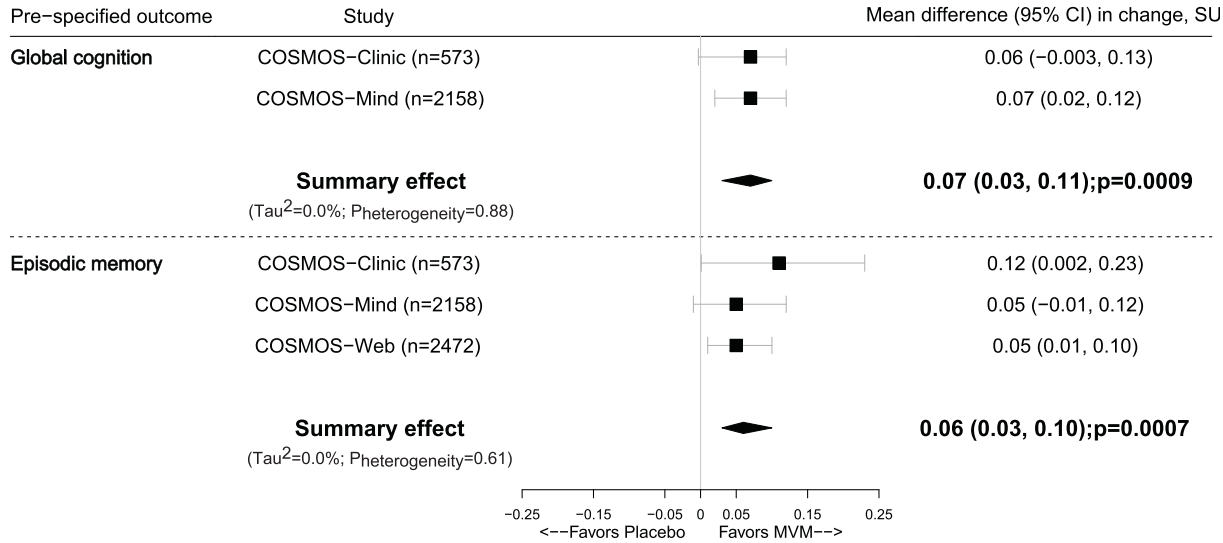

FIGURE 3. Meta-analyses of findings of global cognition and episodic memory, comparing MVM and placebo groups, among nonoverlapping participants across COSMOS cognitive substudies.[1] CI, confidence interval; COSMOS, COcoa Supplement and Multivitamin Outcomes Study [14–16]; MVM, multivitamin-mineral; SU, SD units.

[1]Meta-analysis was performed using random-effects models. A pooled summary effect estimate with 95% (CI) was reported. Heterogeneity across substudies (using the Q statistic) and between-study variance [using tau squared (τ^2)] were examined using the %METAANAL macro (DerSimonian–Laird method) [31, 32]. $P < 0.025$ was used to indicate statistical significance for each coprimary outcome after Bonferroni correction [33].

cognitive benefits associated with MVM supplementation over the first 2–3 y of follow-up, observed in 3 COSMOS cognitive substudies, might have been missed in PHS II. Third, differences may be explained by subtle distinctions in the MVM formulations (Centrum Silver) from PHS II to COSMOS (Supplemental Table 2), particularly with the addition of lutein and lycopene for COSMOS.

Study strengths should be noted: well-characterized sample; detailed in-person neuropsychological assessments that assessed subtle changes across multiple cognitive domains; high adherence and retention rates. Furthermore, this report makes a unique contribution by including a meta-analysis of over 5000 COSMOS participants from 3 cognitive substudies randomly assigned to an MVM supplement compared with placebo with prerandomization and follow-up assessments, providing clear evidence for cognitive benefits of MVM supplementation among older adults.

There were also study limitations. First, COSMOS had relatively low racial and ethnic diversity, which may limit generalizability. Second, we are unable to determine whether specific individual vitamins or minerals of the MVM supplement may explain our observed cognitive benefits. Furthermore, we do not know whether these favorable results associated with the COSMOS MVM supplement (Centrum Silver) can be generalized to other broad-based MVM supplements, as our findings may or may not represent a class effect. Third, we did not control for type-1 error in secondary outcomes and subgroup analyses in COSMOS-Clinic; these findings should be considered exploratory. Fourth, although useful for clinical translation, a simplified approach to addressing the clinical relevance of daily MVM in terms of years of aging has inherent limitations.

In conclusion, among 573 COSMOS-Clinic subcohort participants who underwent repeat in-person, detailed neuropsychological assessments over 2 y, MVM supplementation led to a statistically significantly more favorable change in episodic memory compared with placebo. A meta-analysis of 3 COSMOS cognitive substudies ($n >$ 5000) provided clear evidence that daily MVM use showed benefits for both global cognition and memory among older adults; the magnitude of effect on global cognition appears equivalent to reducing cognitive aging by 2 y. Our findings warrant consideration by clinical guidelines committees for the role of daily MVM use in preventing cognitive decline among older adults.

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Author contributions

The authors' responsibilities were as follows – CMV, OIO: had full access to all of the data in the study and took responsibility for the integrity of the data and the accuracy of the data analysis; CMV, JEM, HDS, OIO: concept and design; all authors: acquisition, analysis, or interpretation of data; CMV: initial drafting of the manuscript; all authors: critical revision of the manuscript for important intellectual content; CMV, JEM, NRC, OIO: statistical analysis; JEM, HDS: obtained funding; and JEM, HDS, NRC, OIO: supervision.

Conflict of interest

CMV has received research support from Nestlé-Purina Petcare Company, Mars Edge, and American Foundation for Suicide Prevention. JEM has received research support from Mars Edge. HDS has received research support from Mars Edge, Pure Encapsulations, and Pfizer Inc., and honoraria and/or travel for lectures from the Council for Responsible Nutrition, BASF, NIH, and American Society of Nutrition during the conduct of the study. AMB has received research support from Mars Edge. OIO receives royalties from Springer Publishing for a book on late-life depression prevention. OIO has received research support from the NIH and the Alzheimer's Association. LDB received funding for COSMOS-Mind from the NIH. MAE receives research

support from the NIH and the Alzheimer's Association. MAE is also compensated for service on a Steering Committee for a study conducted by Nestle. No other authors have disclosures to report.

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Data availability

Data described in the manuscript, code book, and analytic code will be made available upon request. The dataset(s) will be de-identified before release for sharing. We will make the data and associated documentation available to users only under a data-sharing agreement. Details on the availability of the study data to other investigators will be on our study website at https://cosmostrial.org/.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ajcnut.2023.12.011.

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